The Sum Discrepancy Algorithm

The "Sum Discrepancy Algorithm" is an iterative (or online) approach to constructing a low-discrepancy coloring for a set system. Its core idea is to process elements one by one and, at each step, assign a color that minimizes a specific global objective function, typically the sum of the current discrepancies of all sets, weighted to prioritize the most imbalanced ones.

1 Formal Definition

Given a finite set system $(X, \mathcal{R})$ where $|X| = n$ and $|\mathcal{R}| = m$. The algorithm processes the elements of X in some order $v_1, v_2, \ldots, v_n$. At the t-th iteration, it determines the color for v_t using only the information from the previously colored elements $\{v_1, \ldots, v_{t-1}\}$.

The decision rule for the color $\chi(v_t) \in \{-1, +1\}$ is defined as follows:

1. **Compute Current Discrepancies**: For every set $S \in \mathcal{R}$, calculate its discrepancy $\text{disc}^t(S)$ based on the colors assigned so far.
2. **Define a Weight Function**: Apply a weight function to these discrepancies. Common choices include:
   - **Power Function**: $\sum_{S \in \mathcal{R}} (\text{disc}^t(S))^c$ for some constant c.
   - **Exponential Function**: $\sum_{S \in \mathcal{R}} \exp(\eta \cdot \text{disc}^t(S))$ for some scaling parameter $\eta > 0$.
3. **Minimize the Objective**: Assign v_t the color (either $+1$ or -1) that minimizes the value of the chosen weight function after this assignment is hypothetically made.

In essence, the algorithm chooses the color that leads to the smallest increase in a weighted sum of all set discrepancies, where sets with larger discrepancies contribute more heavily to the sum.

2 Purpose

The primary goal of this algorithm is to provide a constructive method for finding colorings with low discrepancy. Unlike purely existential proofs (like Spencer's theorem), this algorithm is designed to be efficient and online, meaning it can be implemented to produce a coloring without needing to know the entire set system or the future elements in advance. By optimizing a global potential function (the sum of weighted discrepancies) rather than greedily fixing only the most imbalanced set, it aims to avoid the pitfalls of simpler greedy strategies and achieve bounds like $O(\sqrt{n \log m})$.

If you'd like to explore the analysis of such algorithms in more detail, just let me know.